

Assignment 01
MGT 3202: Engineering Management
Summer 2025-2026

Total Marks: 15; Submission Deadline: Monday, 20th July 2026

Answer all the following questions:

Question 01: Choose the correct answer from the following questions. **(1 Mark)**

- a) A company introduces a new production layout that reduces material movement and waiting time. This initiative mainly aims to improve:
i) Effectiveness ii) Efficiency iii) Innovation iv) Forecast accuracy
- b) Which managerial function is primarily responsible for assigning employees to departments and defining reporting relationships?
i) Planning ii) Organizing iii) Leading iv) Controlling
- c) A company launches Bangladesh's first AI-powered rice cooker. This is primarily an example of
i) Cost competitiveness ii) Innovation iii) Speed iv) Productivity
- d) Demand for air conditioners increases every summer and decreases every winter. This demand pattern is known as:
i) Trend ii) Seasonality iii) Random variation iv) Cycle
- e) A company produces 2,000 bottles using the same number of labor hours that previously produced only 1,800 bottles. This indicates improved
i) Quality ii) Labor Productivity iii) Forecast Accuracy iv) Capacity

Question 02: State whether the following statements are True or False. If a statement is False, write the correct statement. **(1 Mark)**

- a) Improving product quality without considering customer needs always improves competitiveness.
- b) Top-level managers generally require stronger conceptual skills than first-line managers.
- c) A company can improve labor productivity even if its total production remains unchanged.
- d) Forecast accuracy generally decreases as the forecasting horizon becomes longer.
- e) Increasing the selling price of a product always increases its physical productivity.
- f) Random variation in demand usually follows a predictable repeating pattern.

Question 03: Answer the following questions briefly. **(1 Mark)**

- a) Vertex Furniture Ltd. manufactures office desks. To reduce production costs, the company replaces manual operations with automated cutting machines while maintaining the same product quality. Which aspect of competitiveness is Vertex Furniture Ltd. primarily focusing on? Explain briefly.
- b) A mobile phone company estimates next year's demand before deciding how many workers to recruit and how much raw material to purchase. Why is forecasting important in this situation? Explain any two reasons.
- c) Eco-Home Ltd. manufactures LED bulbs. Last year, the company produced 10,000 bulbs using 20 workers. This year, it produced the same number of bulbs using only 16 workers. Has the company's efficiency, effectiveness, or productivity improved? Justify your answer.

- d) Alpha Electronics Ltd. introduces a smart quality inspection system that reduces defective products without increasing production volume. How can this decision improve both productivity and competitiveness? Explain briefly.

Question 04: Cool-Tech Ltd. manufactures split-type air conditioners. The factory operates 2 shifts per day, with 8 working hours per shift, 5 working days per week, and 50 working weeks per year. Each shift employs 18 workers, and the labor wage is \$15 per hour. During 2025, the company incurred \$120,000 in material cost and \$45,000 in energy cost. The annual overhead cost was 20% of the total labor cost. The factory produced 4,800 air conditioners, but 4% failed the final quality inspection. The approved products were sold at an average price of \$450 each. To improve quality, management plans to invest \$10,000 in an automated inspection system for 2026, which is expected to eliminate all defective products while maintaining the same production volume. **(1 Mark)**

- Calculate the multifactor productivity for 2025 in terms of revenue generated per dollar of input.
- Determine the multifactor productivity after implementing the proposed improvement.
- Calculate the percentage improvement in multifactor productivity.
- As the Operations Manager, would you recommend the 2026 investment? Briefly justify your answer.

Question 05: Comfort Furniture Ltd. manufactures office chairs. The factory operates 3 shifts per day, each consisting of 8 working hours, 6 days per week, and 48 weeks per year. Each shift employs 12 workers. During 2025, the factory produced 92,160 office chairs. In 2026, the company introduced semi-automated assembly equipment. As a result, annual production increased to 100,800 chairs, while the workforce and working schedule remained unchanged.

- Calculate the labor productivity (chairs per labor-hour) for 2025. **(1 Mark)**
- Calculate the labor productivity for 2026.
- Determine the percentage improvement in labor productivity.
- Briefly explain one operational reason that could have contributed to this improvement.

Question 06: The weekly demand for packaged bread is shown below.

(1 Mark)

Week	1	2	3	4	5	6
Demand (Loaves)	520	540	535	560	575	570

Management believes that recent demand should have greater influence on future forecasts.

- Estimate the demand for Week 7 using the 3-week Moving Average.
- Estimate the demand for Week 7 using the 3-period Weighted Moving Average, assuming weights of 0.50, 0.30, and 0.20 for the most recent observations.
- Which forecasting method is expected to respond more quickly to sudden changes in demand? Explain briefly.

Question 07: The monthly demand for microwave ovens is shown below.

(1 Mark)

Month	1	2	3	4	5	6
Demand (Units)	460	475	470	490	485	500

Use Exponential Smoothing with $\alpha = 0.30$ to estimate the forecast for Month 7. (Start with Naïve method to determine the forecast for Month 2 and then proceed with exponential smoothing for months 3 to 7)

Question 08: A consulting firm employs three analysts, each working eight hours per day. Each analyst completes an average of six client reports per day. The firm's daily payroll cost for the analysts is \$900, and the daily overhead expense is \$450.

The firm is considering implementing a new project management tool that should allow each analyst to complete eight reports per day. However, the daily overhead expense will increase to \$500. **(1 Mark)**

Calculate:

- The labor productivity before & after implementing the new project management tool (in reports completed per labor hour)
- The multifactor productivity before & after implementing the new project management tool (in reports completed per dollar of input)

Question 09: Tech-Track Corporation sells smart sensors used in industrial automation. Monthly sales for a seven-month period are as follows:

Month	Sales (in thousands)
January	25
February	27
March	24
April	26
May	28
June	30
July	29

(2 Marks)

Forecast August sales volume using each of the following methods:

- The naïve approach
- A four-month moving average
- A weighted average using 0.60 for July, 0.30 for June, and 0.10 for May
- Exponential smoothing with a smoothing constant equal to 0.25 (Use naïve method to assume February forecast and proceed with exponential smoothing).

Question 10: Two different forecasting techniques (F1 and F2) were used to predict the demand for cases of bottled water. The actual demand and the two sets of forecasts are provided below: **(1 Mark)**

Period	Actual Demand	Forecasted Demand	
		F1	F2
1	68	66	66
2	75	68	68
3	70	72	70
4	74	71	72
5	69	72	74
6	72	70	76
7	80	71	78
8	78	74	80

- Compute MAD, MSE and MAPE for each set of forecasts.
- Based on the results, which forecasting technique appears to be more accurate? Provide a brief explanation.

Question 11: Tech World Electronics is a popular local store that specializes in selling the latest smartphones. The store monitors weekly sales to accurately forecast demand and manage its inventory. The store manager, Ms. Khadiza, needs to forecast smartphone sales for Week 7, Week 8, and Week 9 based on the previous sales data below: **(2 Marks)**

Week	Sales (Units) (in thousands)
Week 1	500
Week 2	450
Week 3	480
Week 4	530
Week 5	550
Week 6	510

- Forecast the sales for Week 5, 6, and 7 using a three-week moving average.
- Forecast the sales for Week 7 using exponential smoothing with an $\alpha = 0.2$. (Hint: Use the naïve method to assume the forecasted value for Week 2)
- If the actual sales for Week 7 turn out to be 535,000 units, compare the forecasting methods (from a and b above) using Mean Absolute Deviation (MAD) and Mean Squared Error (MSE). Determine which method provides a better forecast. (Consider Weeks 5, 6, and 7 for comparison)

Question 12: The monthly sales of electric bicycle sales are shown below.

(1 Mark)

Month	1	2	3	4	5	6
Sales	420	445	460	490	505	530

The factory currently has a production capacity of 560 bicycles per month. Management will expand production if the forecasted demand exceeds the available capacity.

- Develop the linear trend equation using the least squares method.
- Forecast the sales for Months 7 and 8.
- Based on your forecasts, determine whether the company should expand its production capacity.

Question 13: Two forecasting techniques were tested before selecting one for production planning.

(1 Mark)

Month	Actual	Method A	Method B
1	520	515	522
2	535	540	531
3	548	544	546
4	560	565	556
5	555	552	558
6	570	567	573

- Calculate the MAD, MSE and MAPE for both forecasting methods.
- Which forecasting method would you recommend? Justify your answer using the calculated measures.

Answer to the Q. No-01

- a) ii) Efficiency
- b) ii) Organizing
- c) ii) Innovation
- d) ii) Seasonality
- e) ii) Labor Productivity

Answer to the Q. No-02

- a) False $\Rightarrow$ Improving product quality in ways that satisfy customer wants and need can improve competitiveness.
- b) True.
- c) True
- d) True
- e) False $\Rightarrow$ Increasing the selling price may increase revenue based productivity, but it does not necessarily increase physical productivity.
- f) False $\Rightarrow$ Random variation consists of unpredictable residual variations remaining after other line-level patterns are considered.

Answer to the Q. No-03

a) Venter Furniture Ltd. is primarily focusing on cost competitiveness.

By replacing manual operations with automated cutting machines, the company can reduce production cost while maintaining the same level of product quality.

b) Forecasting is important because it helps the mobile phone company estimate future demand and make appropriate operational decisions. Here we can discuss two reasons:

Reason-1: Workforce planning

⇒ If demand is expected to increase, the company can recruit and train the required number of workers in advance.

⇒ If demand is expected to decrease, the company can avoid unnecessary recruitment and labor costs.

Reason-2: Raw material purchasing

⇒ A demand forecast helps to determine how much raw material should be purchased.

c) Given,

For the last year,

$$\begin{aligned} \text{output} &= 10,000 \text{ bulbs} \\ \text{workers} &= 20 \end{aligned} \Rightarrow \text{productivity} = \frac{10,000}{20} = 500$$

For this year,

$$\begin{aligned} \text{Output} &= 10,000 \text{ bulbs} \\ \text{workers} &= 16 \end{aligned} \Rightarrow \text{productivity} = \frac{10,000}{16} = 625$$

$$\begin{aligned} \text{productivity improvement} &= \frac{625 - 500}{500} \times 100\% \\ &= 25\% \end{aligned}$$

So, productivity and efficiency improved. But effectiveness has unchanged, because total output remains same.

d) When defective products decrease, the quality of approval or saleable output produced from the same amount of input increases.

$$\text{productivity} = \frac{\text{Good output}}{\text{Input}}$$

Therefore, it increases good output from the same input, lowers per-unit cost and improves customer satisfaction, productivity and quality competitiveness.

_____ x _____

Answer to the Q. No-04

a) Given,

$$\begin{aligned}\text{Quality Rate} &= 100\% - 4\% \\ &= 96\%\end{aligned}$$

Now,

$$\begin{aligned}\text{MFP}_{2025} &= \frac{4800 \times 0.96 \times 150}{2 \times 18 \times 8 \times 5 \times 50 \times 15 + 2 \times 18 \times 8 \times 5 \times 50 \times 15 \times 0.2 \\ &\quad + 120,000 + 15,000} \\ &= \frac{20,73,600}{1080000 + 216000 + 120000 + 15000} \\ &= 1.42 \quad \text{Ans.}\end{aligned}$$

b) Here,

$$\begin{aligned}\text{MFP}_{2026} &= \frac{4800 \times 150}{1080000 + 216000 + 120000 + 15000 + 10000} \\ &= 1.47 \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{c) MFP Improvement} &= \frac{\text{MFP}_{2026} - \text{MFP}_{2025}}{\text{MFP}_{2025}} \times 100 \\ &= \frac{1.47 - 1.42}{1.42} \times 100 \\ &= 3.46\%\end{aligned}$$

d) Here we can see,

$$\text{MFP}_{2026} > \text{MFP}_{2025}$$

The investment should be recommended. Because \$10,000 investment reduces defects and increases MFP by 3.46%.

_____ x _____

Answer to the Q. No-05

a) Labor Productivity for 2025,

$$LP_{2025} = \frac{92,160}{3 \times 12 \times 8 \times 6 \times 48}$$
$$= 1.11$$

Ans.

b) Labor productivity for 2026,

$$LP_{2026} = \frac{100800}{3 \times 12 \times 8 \times 6 \times 48}$$
$$= 1.22$$

Ans.

c) Here,

$$\text{Improvement} = \frac{LP_{2026} - LP_{2025}}{LP_{2025}} \times 100$$
$$= \frac{1.22 - 1.11}{1.11} \times 100$$
$$= 9.91\%$$

d) Semi-automated assembly equipment can increase assembly speed by reducing manual processing and waiting time. As a result, it has been possible to produce more chairs using the same workforce and working hours.

_____ x _____

Answer to the Q. No - 06

a) Three week moving average,

$$F_7 = \frac{570 + 575 + 560}{3}$$

$$= 568.33$$

Ans.

b) Weighted Moving Average,

$$F_7 = 570 \times 0.5 + 575 \times 0.3 + 560 \times 0.2$$

$$= 569.5$$

Ans.

c) Weighted moving average method is expected to respond more quickly to sudden changes in demand. Because that gives more weight to recent demand. So, if demand changes suddenly, it reacts faster than simple moving average.

Answer to the Q. No - 07

We know, $F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$ Given, $\alpha = 0.3$
 $F_7 = ?$

Month	Demand (units)	Forecasting
1	460	
2	475	460
3	470	$460 + 0.3(475 - 460) = 464.5$
4	490	$464.5 + 0.3(470 - 464.5) = 466.15$
5	485	$466.15 + 0.3(485 - 466.15) = 473.305$
6	500	$473.305 + 0.3(500 - 473.305) = 476.81$

Now for month x ,

$$F_x = 476.81 + 0.3 (500 - 476.81) \\ = 483.77 \text{ units.}$$

————— x —————

Answer to the Q. No-08

a) Labor productivity before,

$$LP_{\text{before}} = \frac{3 \times 6}{3 \times 8} \\ = 0.75$$

After implementing the tool, Labor productivity after,

$$LP_{\text{after}} = \frac{3 \times 8}{3 \times 8} \\ = 1$$

b) Labor productivity before,

$$\begin{aligned} \text{MFP}_{\text{before}} &= \frac{3 \times 6}{900 + 450} \\ &= 0.0133 \end{aligned}$$

After implementing the tool, Labor productivity after,

$$\begin{aligned} \text{MFP}_{\text{after}} &= \frac{3 \times 8}{900 + 500} \\ &= 0.017 \end{aligned}$$

Labor productivity increased from 0.75 to 1 reported per labor-hour, while MFP increased from 0.0133 to 0.017 reported per dollar.

Answer to the Q. No-09

a) Following the naive approach,

$$\begin{aligned}F_{\text{August}} &= A_{\text{July}} \\&= 29 \text{ thousand. Ans.}\end{aligned}$$

b) Four month moving average,

$$\begin{aligned}F_{\text{August}} &= \frac{29 + 30 + 28 + 26}{4} \\&= 28.25 \text{ thousand Ans.}\end{aligned}$$

c) Weighted Average,

$$\begin{aligned}F_{\text{August}} &= 29 \times 0.6 + 30 \times 0.3 + 28 \times 0.1 \\&= 29.2 \text{ thousand Ans.}\end{aligned}$$

d) Given,

$$\alpha = 0.25$$

Month	Sales	Forecast
January	25	
February	27	25
March	24	25.5
April	26	25.125
May	28	25.344
June	30	26.007
July	29	27.006

Now,

$$\begin{aligned}F_{\text{March}} &= 25 + 0.25(27 - 25) \\&= 25.5 \text{ thousand.}\end{aligned}$$

$$\begin{aligned}F_{\text{April}} &= 25.5 + 0.25(24 - 25.5) \\&= 25.125 \text{ thousand}\end{aligned}$$

As a same way,

$$\begin{aligned}F_{\text{August}} &= 27.006 + 0.5(29 - 27) \\&= 27.504 \text{ thousand}\end{aligned}$$

_____ x _____

Answer to the Q. No-10

For Technique F1,

Period	Actual (A)	F1 (F)	A-F	(A-F) ²	APE
1	68	66	2	4	2.94%
2	75	68	7	49	9.33%
3	70	72	2	4	2.86%
4	74	71	3	9	4.05%
5	69	72	3	9	4.35%
6	72	70	2	4	2.78%
7	80	71	9	81	11.25%
8	78	74	4	16	5.13%
Total			32	176	42.69%

We know,

$$APE = \frac{|A-F|}{A} \times 100$$

Now,

$$MAD = \frac{32}{8} = 4$$

$$MSE = \frac{176}{8-1} = 25.14$$

$$MAPE = \frac{42.69}{8} = 5.34\%$$

For Technique F2,

period	Actual	F2	A-F	(A-F) ²	APE
1	68	66	2	4	2.94%
2	75	68	7	49	9.33%
3	70	70	0	0	0%
4	74	72	2	4	2.7%
5	69	74	5	25	7.25%
6	72	76	4	16	5.56%
7	80	78	2	4	2.5%
8	78	80	2	4	2.56%
Total			24	106	32.84%

Now,

$$MAD = \frac{24}{8} = 3$$

$$MSE = \frac{106}{8-1} = 15.14$$

$$MAPE = \frac{32.84}{8}$$

$$= 4.11\%$$

So, Technique F2 is more accurate.

b) Here we can see,

$$MAD_{F1} > MAD_{F2}$$

$$MSE_{F1} > MAD_{F2}$$

$$MAPE_{F1} > MAPE_{F2}$$

So, Technique F2 is more accurate, because all error measure lower from F1.

————— x —————
Answer to the Q. No - 11

a) Three-week Moving Average

$$\text{for week 5, } F_5 = \frac{450 + 480 + 530}{3} \\ = 486.67$$

$$\text{For week 6, } F_6 = \frac{480 + 530 + 550}{3} \\ = 520$$

$$\text{For week 7, } F_7 = \frac{530 + 550 + 510}{3} \\ = 530$$

b) Given,
 $\alpha = 0.2$

Week	Sales	Forecast
1	500	
2	450	4500
3	480	490
4	530	488
5	550	496.4
6	510	507.12

Here,

$$F_2 = A_1 \\ = 500$$

$$F_3 = 500 + 0.2 \times (450 - 500) \\ = 490$$

And a same way,

$$F_7 = 507.12 + 0.2(510 - 507.12) \\ = 507.70$$

c) Moving Average Error:

Week	Actual	Forecast	$ A-F $	$(A-F)^2$
5	550	486.67	63.33	4011.11
6	510	520	10	100
7	535	530	5	25
Total			78.33	4136.11

$$MAD_{MA} = \frac{78.33}{3} = 26.11$$

$$MSE_{MA} = \frac{4136.11}{3-1} = 2068.06$$

Now For Exponential Smoothing Error:

Week	Actual	Forecast	$ A-F $	$(A-F)^2$
5	550	496.4	53.60	2872.96
6	510	507.12	2.88	8.29
7	535	507.696	27.304	745.51
Total			83.784	3626.76

$$MAD_{ES} = \frac{83.784}{3} = 27.93$$

$$MSE_{ES} = \frac{3626.76}{2} = 1813.38$$

Here we can see,

$$MSE_{MA} > MSE_{ES}$$

So, Exponential smoothing is preferable based on MSE.

_____x_____

Answer to the Q. No - 12

Given,

Month (t)	t^2	Sales (y)	ty
1	1	420	420
2	4	445	890
3	9	460	1380
4	16	490	1960
5	25	505	2525
6	36	530	3180
Total	91	2850	10355

Now,

$$b = \frac{6 \times 10355 - 21 \times 2850}{6 \times 91 - 21^2}$$
$$= 21.7143$$

$$a = \frac{2850 - 21.7143 \times 21}{6}$$
$$= 399$$

a) our trend equation, $F_t = 399 + 21.7143t$

b) For $F_7 = 399 + 21.7143 \times 7$
 $= 551$ bicycles.

$$F_8 = 399 + 21.7143 \times 8$$
$$= 572 \text{ bicycles.}$$

e) Here we can see,
 $F_7 < 560$, $F_8 > 560$

So, the company should expand its production capacity.

Answer to the Q. No-13

a) Accuracy comparison for method A:

Month	Actual (A)	Forecast (F)	A-F	(A-F) ²	APE
1	520	515	5	25	0.96%
2	535	540	5	25	0.93%
3	548	544	4	16	0.73%
4	560	565	5	25	0.89%
5	555	552	3	9	0.54%
6	570	567	3	9	0.53%
Total			25	109	0.59%

Now,

$$MAD_A = \frac{25}{6} = 4.17$$

$$MSE_A = \frac{109}{6-1} = 21.8$$

$$MAPE_A = \frac{4.59}{6} = 0.76\%$$

Accuracy comparison for method B:

Month	Actual (A)	Forecast	A-F	(A-F) ²	APE
1	520	522	2	4	0.38%
2	535	531	4	16	0.75%
3	548	546	2	4	0.37%
4	560	556	4	16	0.71%
5	555	558	3	9	0.54%
6	570	573	3	9	0.53%
Total			18	58	3.28%

Now,

$$MAD_B = \frac{18}{6} = 3$$

$$MSE_B = \frac{58}{6-1} = 11.6$$

$$MAPE_B = \frac{3.28}{6} = 0.55\%$$

b) Here we can see,

$$MAD_A > MAD_B$$

$$MSE_A > MSE_B$$

$$MAPE_A > MAPE_B$$

So, Method B is more accurate, because all error measure lower from method A.

————— x —————